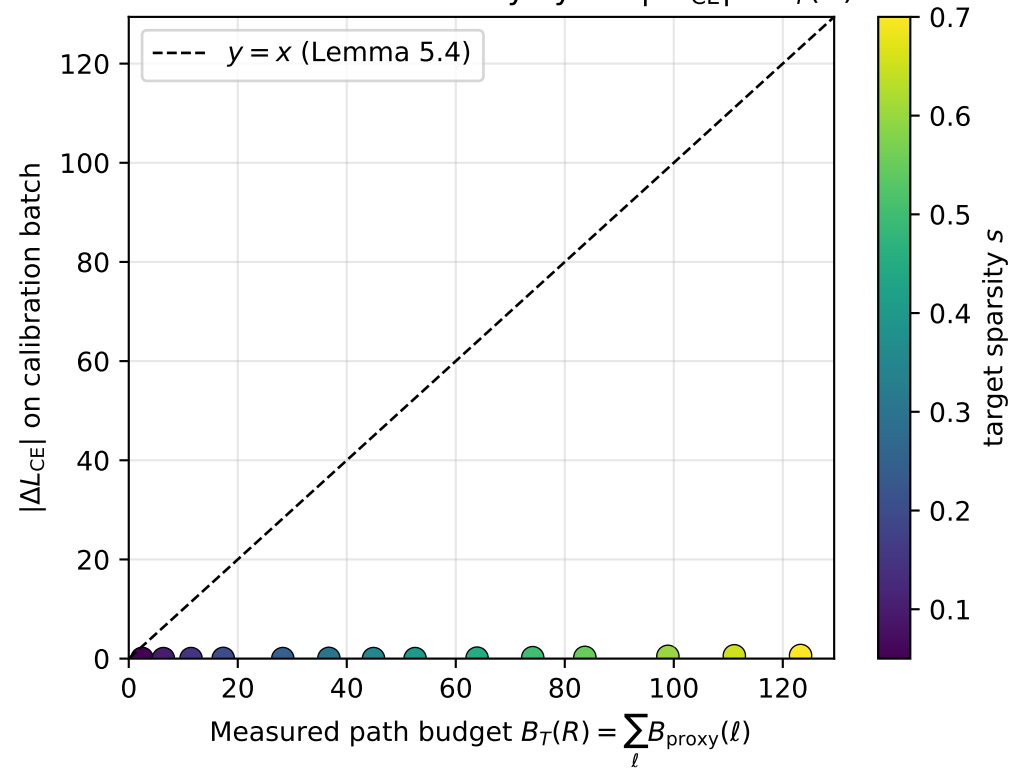


Lemma 5.4 holds at every cycle: $|\Delta L_{CE}| \leq B_T(R)$



Both sides of Lemma 5.4 grow together with sparsity

